

Russell Dudek

Industrial Systems Modernization | Manufacturing Data & Automation

Pittsburgh, Pennsylvania | 412.287.8640

russelldudek@gmail.com | [linkedin.com/in/russelldudek](https://www.linkedin.com/in/russelldudek)

<https://russelldudek.github.io/ByteCubit/>

Do not migrate screens. Prove the signal path.

ACTUAL MANDATE

Safely modernize legacy MES functionality into maintainable Ignition applications while improving SQL and historian context and creating cloud-ready manufacturing data without destabilizing production.

COMPANY AND ROLE CONTEXT

Bytecubit publicly presents itself as an IT solutions and staff-augmentation firm spanning custom software, data/BI, analytics, cloud, infrastructure, and AI/ML. The posting places this engineer inside a client modernization effort rather than a greenfield product build.

The contract runs through year-end with extension potential, so the engineer must create immediate value and leave a repeatable delivery system rather than a collection of bespoke screens.

CENTRAL CONTRADICTION

- Move fast enough to show progress, but slowly enough to protect live manufacturing.
- Standardize applications and data, but preserve legitimate plant/process differences.
- Clean historian context without breaking unknown consumers.
- Expose data to the cloud without weakening OT reliability, ownership, or semantics.

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Signal Twin Workbench

OPERATING MODEL

1. Read the legacy function

Map process states, users, inputs, logic, SQL transactions, historian points, consumers, consequences, and support ownership.

2. Build the target twin

Define Ignition project/component boundaries, tags/UDTs, queries, diagnostics, access, tests, and data context.

3. Reconcile evidence

Compare function parity, exceptions, data quality, timestamps, usability, performance, security, supportability, and rollback.

4. Decide deliberately

Promote, hold, repair, or retire with a named decision owner and visible evidence.

5. Return learning

Capture reusable components, naming rules, tests, runbooks, data contracts, and governance decisions for the next tranche.

WHY THIS MATTERS

Modernization becomes a sequence of governed releases. The organization gains a reusable method for changing applications, historian context, and cloud data without treating production as the test environment.

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BALANCED SCORECARD

Production continuity

Release behavior, downtime/rework categories, rollback readiness, post-release observation.

Application quality

Parity, exception coverage, performance, diagnostic completeness, support ownership.

Data trust

Names, units, timestamps, quality, asset relationships, completeness, consumer map.

Adoption and support

Operator task success, training, runbook quality, issue path, recurring failure visibility.

Reuse and learning

Reusable components, UDT/tag patterns, named queries, tests, conventions, and release records.

Portfolio value

Prioritized next tranche by operational value, dependency risk, readiness, and reuse potential.

EVIDENCE MAP

- Vape-Jet: direct factory, machine-fleet, telemetry, support, ERP, quality, and engineering issue-flow integration.
- Compunetics: advanced electronics, high-reliability manufacturing, quality systems, process controls, and technical programs.
- Amazon: 24/7 launch operations, standard work, throughput visibility, escalation, and customer commitments.
- DudeWorth: AI/automation workflow design, structured context, human review, governance, evaluation, and adoption.

EVIDENCE BOUNDARY

No production Ignition tenure or AVEVA PI administration is claimed. Platform fit should be evaluated through a bounded representative working session.

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EXECUTIVE INTERVIEW QUESTIONS

- Which production function creates the most costly recurring friction today, and how is that cost observed?
- What percentage of the modernization burden is application development versus dependency discovery, data cleanup, and support ownership?
- How are Ignition projects, tags/UDTs, queries, scripts, environments, credentials, release, and rollback governed today?
- Which PI points, AF structures, SQL jobs, reports, and cloud consumers are known, and where does tribal knowledge remain?
- Who has authority to accept behavior, data semantics, production release, rollback, and ongoing support?
- What would make the client extend the contract at year-end: coding throughput, released operational value, reusable standards, data quality, or reduced support burden?

FIRST 90 DAYS

Days 1-30

Read the system, map stakeholders/dependencies, triage immediate risks, select a bounded pilot, and define the acceptance method.

Days 31-60

Build the target twin, reconcile application and data behavior, and prepare cutover/rollback evidence.

Days 61-90

Release or hold deliberately, observe production, publish reusable patterns, and prioritize the next modernization tranche.

CANDIDATE THESIS

The most valuable engineer is not merely the person who can configure the next screen. It is the person who can make the signal path, release evidence, data context, and support ownership explicit enough that modernization compounds safely.